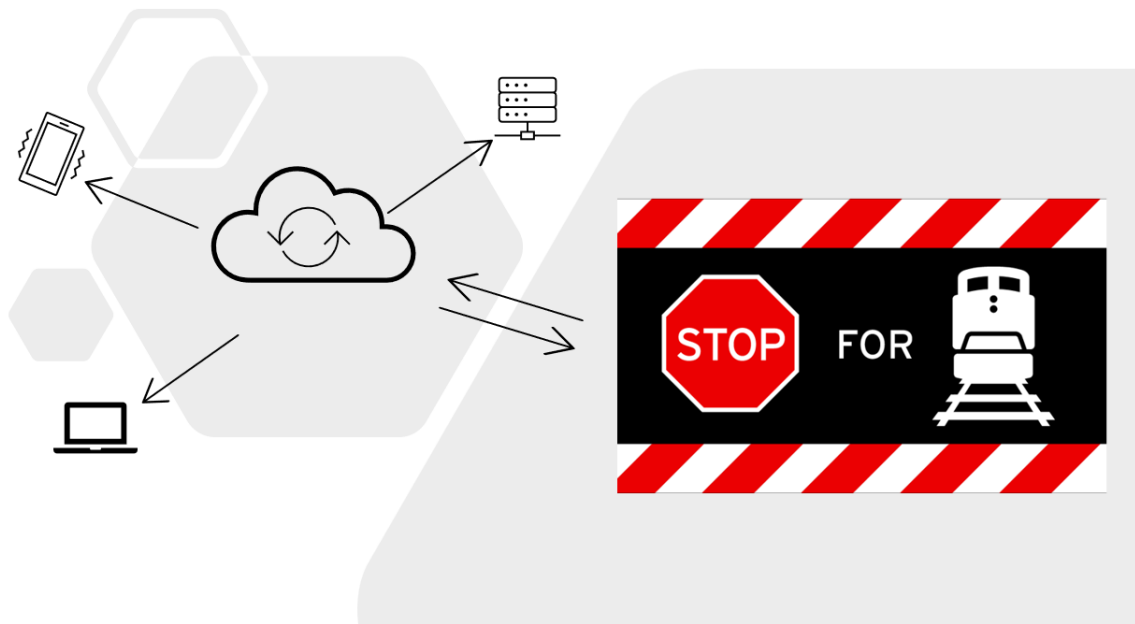




U.S. Department
of Transportation
**Federal Railroad
Administration**

Office of Research,
Development and Technology
Washington, DC 20590

Rail Crossing Violation Warning Application Phase III



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REPORT DOCUMENTATION PAGE					Form Approved OMB No. 0704-0188	
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1. REPORT DATE (DD-MM-YYYY) 7-03-2025		2. REPORT TYPE		3. DATES COVERED (From - To) September 2022–March 2025		
4. TITLE AND SUBTITLE Rail Crossing Violation Warning Application – Phase II				5a. CONTRACT NUMBER 693JJ6-22-C-000037		
				5b. GRANT NUMBER		
				5c. PROGRAM ELEMENT NUMBER		
6. AUTHOR(S) Alejandro Sanchez-Badillo 0000-0003-0454-0488 , Gregory Baumgardner 0000-0002-2211-9876 ,				5d. PROJECT NUMBER		
				5e. TASK NUMBER		
				5f. WORK UNIT NUMBER		
7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES) Battelle Memorial Institute 505 King Ave Columbus, OH 43201				8. PERFORMING ORGANIZATION REPORT NUMBER		
9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES) U.S. Department of Transportation Federal Railroad Administration Office of Railroad Policy and Development Office of Research, Development and Technology Washington, DC 20590				10. SPONSOR/MONITOR'S ACRONYM(S)		
				11. SPONSOR/MONITOR'S REPORT NUMBER(S)		
12. DISTRIBUTION/AVAILABILITY STATEMENT						
13. SUPPLEMENTARY NOTES COR: Jared Withers						
14. ABSTRACT This report summarizes the Rail-Crossing Violation Warning Phase III (RCVW III) project from September 2022 to March 2025. This project builds upon prior development work by Battelle for the Federal Railroad Administration (FRA) RCVW III enhances the RCVW capabilities by adding a Cloud-based Subsystem and updating its Connected Vehicle technology to align with the Federal Communications Commission C-V2X Auto Safety Spectrum Rules.						
15. SUBJECT TERMS Rail Crossing Violation Warning, RCVW, connected vehicles, CV, Connected Automated Vehicles, CAV, Highway-Rail Intersection, Dedicated Short Range Communications, DSRC, signal phase and timing, SPaT, Map Data, MAP, Institute of Electrical and Electronics Engineers, IEEE, C-V2X. Cloud Messaging						
16. SECURITY CLASSIFICATION OF:			17. LIMITATION OF ABSTRACT	18. NUMBER OF PAGES 52	19a. NAME OF RESPONSIBLE PERSON	
a. REPORT	b. ABSTRACT	c. THIS PAGE			19b. TELEPHONE NUMBER (Include area code)	

METRIC/ENGLISH CONVERSION FACTORS

ENGLISH TO METRIC

LENGTH (APPROXIMATE)

1 inch (in)	=	2.5 centimeters (cm)
1 foot (ft)	=	30 centimeters (cm)
1 yard (yd)	=	0.9 meter (m)
1 mile (mi)	=	1.6 kilometers (km)

AREA (APPROXIMATE)

1 square inch (sq in, in ²)	=	6.5 square centimeters (cm ²)
1 square foot (sq ft, ft ²)	=	0.09 square meter (m ²)
1 square yard (sq yd, yd ²)	=	0.8 square meter (m ²)
1 square mile (sq mi, mi ²)	=	2.6 square kilometers (km ²)
1 acre = 0.4 hectare (he)	=	4,000 square meters (m ²)

MASS - WEIGHT (APPROXIMATE)

1 ounce (oz)	=	28 grams (gm)
1 pound (lb)	=	0.45 kilogram (kg)
1 short ton = 2,000 pounds (lb)	=	0.9 tonne (t)

VOLUME (APPROXIMATE)

1 teaspoon (tsp)	=	5 milliliters (ml)
1 tablespoon (tbsp)	=	15 milliliters (ml)
1 fluid ounce (fl oz)	=	30 milliliters (ml)
1 cup (c)	=	0.24 liter (l)
1 pint (pt)	=	0.47 liter (l)
1 quart (qt)	=	0.96 liter (l)
1 gallon (gal)	=	3.8 liters (l)
1 cubic foot (cu ft, ft ³)	=	0.03 cubic meter (m ³)
1 cubic yard (cu yd, yd ³)	=	0.76 cubic meter (m ³)

TEMPERATURE (EXACT)

$$[(x-32)(5/9)]^{\circ}\text{F} = y^{\circ}\text{C}$$

METRIC TO ENGLISH

LENGTH (APPROXIMATE)

1 millimeter (mm)	=	0.04 inch (in)
1 centimeter (cm)	=	0.4 inch (in)
1 meter (m)	=	3.3 feet (ft)
1 meter (m)	=	1.1 yards (yd)
1 kilometer (km)	=	0.6 mile (mi)

AREA (APPROXIMATE)

1 square centimeter (cm ²)	=	0.16 square inch (sq in, in ²)
1 square meter (m ²)	=	1.2 square yards (sq yd, yd ²)
1 square kilometer (km ²)	=	0.4 square mile (sq mi, mi ²)
10,000 square meters (m ²)	=	1 hectare (ha) = 2.5 acres

MASS - WEIGHT (APPROXIMATE)

1 gram (gm)	=	0.036 ounce (oz)
1 kilogram (kg)	=	2.2 pounds (lb)
1 tonne (t)	=	1,000 kilograms (kg)
	=	1.1 short tons

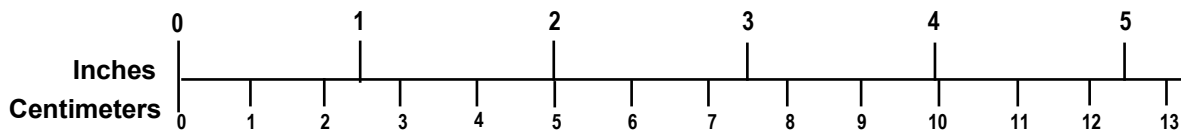
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1 milliliter (ml)	=	0.03 fluid ounce (fl oz)
1 liter (l)	=	2.1 pints (pt)
1 liter (l)	=	1.06 quarts (qt)
1 liter (l)	=	0.26 gallon (gal)
1 cubic meter (m ³)	=	36 cubic feet (cu ft, ft ³)
1 cubic meter (m ³)	=	1.3 cubic yards (cu yd, yd ³)

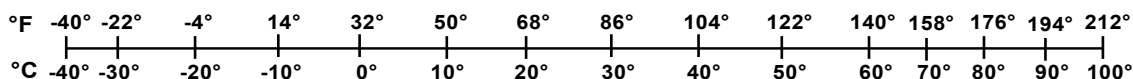
TEMPERATURE (EXACT)

$$[(9/5)y + 32]^{\circ}\text{C} = x^{\circ}\text{F}$$

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QUICK FAHRENHEIT - CELSIUS TEMPERATURE CONVERSION



For more exact and or other conversion factors, see NIST Miscellaneous Publication 286, Units of Weights and Measures.
Price \$2.50 SD Catalog No. C13 10286

Updated 6/17/98

Acknowledgements

The Battelle team thanks the following organizations for their invaluable support throughout this project.

- Florida Department of Transportation
- Trimble
- Volpe National Transportation Systems Center

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1 Executive Summary

This report summarizes the Rail-Crossing Violation Warning Phase III (RCVW III) project from September 2022 to March 2025. This project builds upon prior development work by Battelle for the Federal Railroad Administration (FRA).

RCVW III enhances the RCVW capabilities by adding a Cloud-based Subsystem (CBS) and updating its Connected Vehicle (CV) technology to align with the Federal Communications Commission C-V2X Auto Safety Spectrum Rules.

The inclusion of the CBS to the RCVW expands its capabilities by allowing the broadcast of active HRI warning devices activation status over the internet to any authorized subscriber.

Results from the data collected during a demonstration of the CBS showed that several train detections were missed and the integrity and accuracy of the data used to compare the train detections was not performed. To obtain an accurate comparison, detections need to be collected in a controlled environment where the integrity of the data is verifiable.

Based on the results of a limited Field Test and Evaluation event performed as part of this project, the update of the CV communication radios to a C-V2X technology did not have a noticeable effect on the functionality of the system and the performance was on par with previous experiences.

This project provided the necessary data to confirm that the RCVW CBS:

- can inform remote users of the activation status of HRI active warning devices.
- is capable of interfacing in multiple ways with traditional traffic management devices in the field to obtain the activation status of HRI active warning devices.
- is capable of storing HRI active warning device activation state changes events as well as static HRI data which can be retrieved for analysis
- does not require a direct connection to Railroad crossing control equipment to obtain HRI active warning devices activation status.
- does not require the installation of additional train sensing equipment
- seamlessly interface with existing DOT and municipality traffic equipment with no disruption to the traffic management activities.

The next logical step is to conduct a rigorous test and verification of the system requirements to fully evaluate the system changes. Once this is complete, an outreach campaign to different RCVW stakeholders is recommended to promote the updated capabilities of the RCVW and generate industry awareness. These stakeholders include local and State agencies, vehicle manufacturers and railway companies. With their support, model deployments and pilot installations across the United States can be performed to evaluate the efficacy of the system in real life scenarios.

2 Introduction

The U.S. Department of Transportation (U.S. DOT) is committed to the research and development of innovative technologies intended on making travel safer. Specifically, the connected vehicle (CV) initiative seeks to create applications and prototypes that rely on the exchange of safety-critical information from vehicle to vehicle (V2V) and between vehicle and infrastructure (V2I) using CV radios. Beginning in 2015, the Federal Railroad Administration (FRA) and its subcontractors have been effective in producing a CV prototype called the Rail-Crossing Violation Warning (RCVW), which is designed to notify approaching drivers of an active rail crossing and warn them of a potential collision.

This report presents the results of the FRA funded RCVW Phase III (RCVW III) research project which builds upon the prior work performed by Battelle for FRA. The main objectives of RCVW-III are:

- update the Data Communication Radios (DCR) of the CV system to Cellular Vehicle-to-Everything (C-V2X) technology
- integrate a Cloud-based Subsystem (CBS) that would allow a subscribed user to receive and/or request highway-rail intersections (HRI)-related information over the internet.

2.1 Background

An urgent need exists for additional protections for vehicle drivers traversing HRIs beyond traditional active warning devices. Table 1 presents a review of FRA safety statistics since 2015¹ that does not reflect appreciable improvement in HRI incidents involving motor vehicles (i.e., excluding pedestrian and other categories), fatalities, or injuries.

Table 1. FRA Safety Statistics at HRIs from Calendar Years 2018–2024

Calendar Year	Motor Vehicle Incident	Fatalities	Injuries
2018	2,029	177	780
2019	2,041	193	780
2020	1,728	124	611
2021	2,010	155	657
2022	2,052	194	813
2023	2,002	152	706

¹ Source: Federal Railroad Administration Office of Safety Analysis [website](#)

Calendar Year	Motor Vehicle Incident	Fatalities	Injuries
2024 ²	1,670	120	531

While active warning devices such as flashing lights, bells, and gates are effective at decreasing risk, these HRIs still account for roughly one-half of all crashes. The statistics also show that the top causes attributed to HRI crashes are distracted drivers and driver judgement errors.

Distracted drivers may not notice they are approaching an HRI, or fail to perceive activated warning devices, or not recognize that a train is approaching. Situational awareness of a vehicle operator may be challenged due to adverse atmospheric conditions, engagement with a myriad of distractions (e.g., cell phone use or personal interactions), the influence of alcohol/drugs/medication, or impaired mental capacity due to fatigue or medical events. Existing warning devices do not communicate with roadway-vehicle systems.

Development of the RCVW application attempts to address this operational safety risk by enhancing the situational awareness of roadway-vehicle drivers when approaching or stopped within an active HRI by using in-vehicle multi-sensory warnings and alerts. In addition, RCVW includes a CBS which provides HRI information such as timely active warning devices status (flashers, gates, bells activated/deactivated), as well as FRA related HRI data over the internet to authorized subscribers.

2.2 Objectives

RCVW Phase I demonstrated a V2I-based technology intended for use at HRIs equipped with active warning devices . When a CV equipped with the RCVW system approaches an RCVW equipped HRI, the system will notify the driver of the upcoming crossing. If the HRI warning devices activate as the vehicle is approaching, the Vehicle-Based Subsystem (VBS) will issue an audible and visual alert to the driver. If the VBS predict a violation due to driver not taking sufficient action, the VBS will issue an audible and visual warning.

RCVW II made refinements to the software and hardware and transformed the proof-of-concept prototype into a more dependable, accurate and widely applicable system. RCVW-III updated the CV communication radios of both the Vehicle-Based Subsystem and Road-Based Subsystem to comply with the latest Federal Communications Commission (FCC) rules, updated the software communications platform and incorporated a CBS to expand the system capabilities.

The following modifications and enhancements were incorporated:

- Development of a CBS within Microsoft Azure.
- Updates of the CV communications radios from Dedicated-Short Range Communications (DSRC) technology to C-V2X.
- Update of both VBS and Roadside-Based Subsystem (RBS) computer platforms (CP).
- Update of the VBS Global Navigation Satellite System (GNSS) device.

²Data for 2024 are preliminary results as the FRA analysis is not yet completed.

- Updates of Driver Vehicle Interface (DVI) visual and audio alerts to produce distinct in-vehicle alerts when a vehicle is occupying the HRI hazard zone based on the status of the HRI active warning devices (activate/not active).

2.3 Overall Approach and scope

The approach was to:

- Acquire a better understanding of the requirements set forth by the FCC in regard to the update and use of the new CV communication radio technology.
- Gain a perspective on the different protocols used to access the Cloud and identify the most robust C/C++ Open-Source message broker client tools
- Select and incorporate the C-V2X Data Communication Radio (DCR) to the system.

These initial tasks facilitated to help better define the updates on functionality for RCVW III.

The scope for this project was defined as:

- Update of the RCVW Concept of Operations (ConOps), Design and Architecture documentation and RCVW System Requirements document.
- Obtain approval by the FCC to broadcast using C-V2X radios.
- Design and develop the CBS and interface to the RBS
- Developing and constructing the updated system hardware and software
- Design and execute a bench test of the CBS
- Design and execute a controlled field test of the updated C-V2X RCVW system
- Design and execute a controlled demonstration of the RCVW CBS

3 Summary of Results and Deliverables

3.1 System Engineering Documentation

The system engineering documentation for the RCVW system was developed during Phase I and Phase II of this project. The updates in the communication technology as well as inclusion of the CBS required this documentation to be updated. These documents are available to the public and can be accessed at the RCVW GitHub repository.³

3.1.1 Concept of Operations Document

The ConOps [1] describes the current state of operations of the RCVW system, establishes the rationale for including a cloud-based component, and describes the RCVW system in terms of its features and operations.

Prior to the RCVW system concept development, the Cooperative Intersection Collision Avoidance Systems (CICAS) initiative formed a working group by partnering with automotive Original Equipment Manufacturers (OEMs), State and local Department of Transportation (DOT)s, and university research centers throughout the country. The goals of that initiative were to:

- Develop and demonstrate CICAS for violations of traffic control devices and to assist a CV in maneuvering through cross traffic
- Assess the value and acceptance of CICAS
- Develop and provide tools to support industry developments

Subsequent to the conclusion of CICAS program, the FRA initiated a project carried out by the Volpe National Transportation Systems Center to leverage the CICAS research to develop a ConOps and System Requirements Specification (SRS) for a conceptual rail collision avoidance system utilizing CV technology. Following the completion of these efforts, the FRA utilized the ConOps and System Requirements Specification (SRS) as the basis for developing a prototype RCVW system.⁴ The RCVW system proved that CV technologies could be employed to provide in-vehicle rail crossing violation warnings. It demonstrated that actionable information could be communicated wirelessly from infrastructure to, and for the use by, an approaching CV. The FRA then proceeded to fund Phase II of the RCVW system, taking it from a proof of concept to a dependable, accurate, and widely applicable system. Phase II enhanced the system reliability and incorporated additional functionality that demonstrated the potential for leveraging real-time CV concepts and services to enhance rail crossing safety.⁵

The effort in this project was to update the ConOps to:

1. expand, enhance, and refine the capabilities of the existing system by including a cloud-based component.

³ <https://github.com/RCVW-PHASEIII/RCVW/tree/master/doc>

⁴ FHWA-JPO-18-609 “Vehicle-to-Infrastructure Prototype Rail Crossing Violation Warning Application. Project Report”

⁵ DOT FRA ORD-22/07 “Rail Crossing Violation Warning Application – Phase II”

2. incorporate C-V2X communications; and
3. Identify scenarios in which the cloud-based component might be used to provide benefits of the RCVW concept to roadway vehicles that do not have CV technology.

The latest version of the ConOps can be accessed at the RCVW GitHub repository⁶.

3.1.2 System Requirements Specification Document

The RCVW SRS [2] document describes the operational requirements for a Vehicle-to-Everything (V2X) RCVW CV safety application enhanced with a cloud-based messaging module.

The document presents the requirements for an HRI connected vehicle safety application that is based on CV and complemented with cloud-computing concepts. The requirements detail how a cloud-based subsystem integrates into the RCVW roadside-based component at HRIs equipped with active warning systems.

The project systems engineers along with the development team revised the existing requirement traceability matrix against the updated user needs presented in the revised ConOps. A major update was the creation of requirements to satisfy the needs to implement the CBS (See Table 2). The addition of this subsystem resulted in updates to several Roadside-based Subsystem requirements as well.

Table 2. CBS Requirements

Req Id	Requirement
CBS-001	The CBS shall maintain 99.99% cloud service availability when the communications link is healthy.
CBS-002	The CBS shall prevent unauthorized cyber access.
CBS-003	The CBS shall employ methods to ensure incoming data is from an authorized RCVW RBS.
CBS-004	The CBS shall implement methods for Cloud based data storage.
CBS-005	All HRI/RBS related information stored within the CBS shall be available for querying when requested by an authorized user
CBS-006	The CBS shall be capable to store HRI related attributes following the guidance of the FRA Data Dictionary for External Use Grade Crossing Inventory System (GCIS)
CBS-007	The CBS shall be capable of storing IEEE-1570 Message Type 4904 data elements to the extent they can be obtained.
CBS-008	Each data entry stored in the CBS shall be linked to a unique HRI ID.
CBS-009	Each HRI ID stored in the CBS shall be linked to a Latitude and longitude attribute

⁶https://github.com/RCVW-PHASEIII/RCVW/blob/master/doc/RCVW-III_ConOps%20Final.pdf

Req Id	Requirement
CBS-010	Each HRI ID stored in the CBS shall be linked to an attribute indicating if an RBS is present
CBS-011	The CBS shall be capable of storing HRI geometry (MAP) for all HRIs linked to an RBS
CBS-012	The CBS shall be capable of storing RBS Diagnostic information for HRI's linked to an RBS.
CBS-013	The RBS diagnostic information shall indicate the operational status of the data connection between the CBS and RBS
CBS-014	The RBS diagnostic information shall indicate the radio transmission rates for messages transmitted between the RBS and VBS
CBS-015	The RBS diagnostic information shall indicate the operational status of the RBS components and overall RBS software modules.
CBS-016	The CBS shall be capable of storing the HRI activation status for all HRIs linked to an RBS
CBS-017	The CBS shall assign an Event Streaming Endpoint for each HRI ID linked to an existing RBS.
CBS-018	When the CBS notes a change in the activation Status of an HRI, the CBS shall write a message to the HRI's Event Streaming Endpoint with the state change information.
CBS-019	When explicitly queried, the CBS shall be capable of reporting an array of known RCVW RBS equipped HRI identifiers contained within the boundaries of a specified location and radius.
CBS-020	The CBS shall be capable of maintaining a continuous connection to a Network Transport of RTCM via Internet Protocol (NTRIP) caster for each of the known RBS locations
CBS-021	The CBS shall be capable of sending RTCM correction messages to the corresponding RBS within 2 seconds of reception of RTCM information from an NTRIP caster
CBS-022	The CBS shall receive RBS messages within 0.5 second of being sent
CBS-023	The CBS shall log RBS messages and the corresponding CBS event messages within 5 seconds of being received
CBS-024	The CBS shall retain a log of RBS messages and corresponding CBS event messages.

In addition to the Cloud-based requirements, the following new capabilities are now reflected in the SRS:

Table 3. Updated RCVW Capabilities

Req Id	Requirement
RBS-022b	The CBS shall maintain 99.99% cloud service availability when the communications link is healthy.
VBS-011d	The CBS shall prevent unauthorized cyber access.
VBS-12a	The CBS shall employ methods to ensure incoming data is from an authorized RCVW RBS.

The complete list of requirements can be found at the final SRS which can be accessed at the RCVW GitHub repository⁷.

3.1.3 Architecture and Design Specification Document

The Architecture and Design Specification [3] provides the specifications for the RCVW CV system as well as the cloud-based messaging system. The document describes the architecture, i.e., the design and integration, of the components that comprise this system including methods of its installation. The following interfaces are defined in the document:

- RBS and VBS
- RBS and the existing HRI controller preemption connection
- RBS and CBS
- CBS and third-party stakeholders – authorized subscribers

The addition of the CBS to the RCVW system architecture resulted in a redesign of the base architecture of the RBS. This change was required for this subsystem to allow the exchange of information with the cloud. Figure 1 shows the updated system architecture⁸.

⁷ [RCVW/doc/RCVW-III_SRS_Final.pdf at master · RCVW-PHASEIII/RCVW · GitHub](#)

⁸ Connection lines in the figure represent data transmission paths.]

The Operating System (OS) selected for the Computing Platforms of the RCVW is Ubuntu Linux as it allows a flexible platform for product development due to it being open source and easily customized. In addition to the OS layer, the RCVW is developed using a set of loosely coupled plugins which will be responsible for registering with the V2I Hub. As part of registration, a plugin will notify the hub which message types it will produce and transmit. Each plugin will also identify which message types it will request to receive from the V2I Hub. Plugins can be either message producers, message consumers or both. A high-level overview diagram of the V2I Hub platform is presented below (Figure 2)

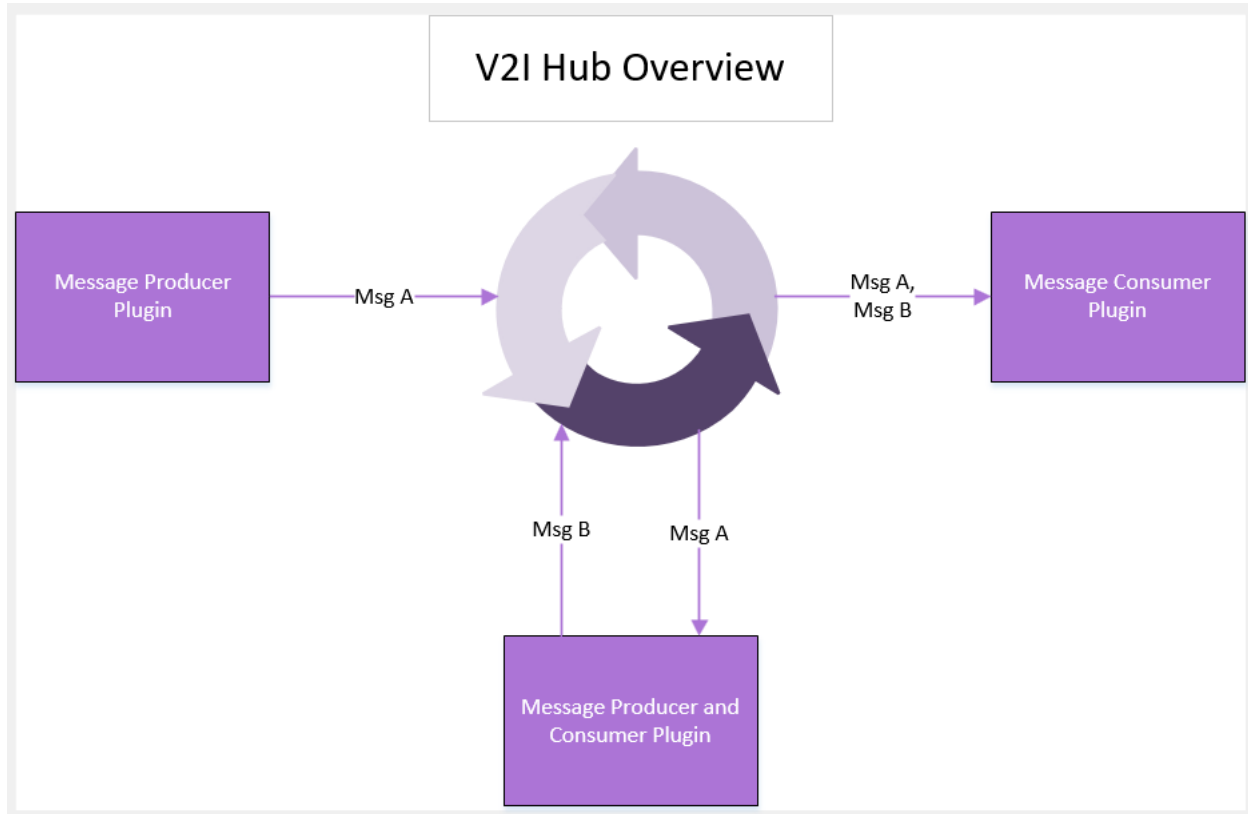


Figure 2. V2I Hub Overview

3.2 Cloud-based Subsystem Development

The addition of the CBS now required to include methods of communication between the RBS and the CBS. The Transmission Control Protocol/Internet Protocol (TCP/IP) backhaul is how the RBS exchanges information with the CBS. It is a low latency high-speed internet connection over ethernet / Wi-Fi. Similar to the communication between the RBS and the VBS, the RBS broadcast both MAP and Signal Phase and Timing (SPaT) messages to the CBS following the SAE J2735 standard. The MAP message contains the intersection geometry including the vehicle lanes and tracked vehicle lanes for the HRI. The SPaT message provides the status of each lane in the MAP message along with the corresponding HRI ID. RBS system diagnostic messages are also transmitted to the CBS as a result of periodic system self-checks. The CBS transmits Radio Technical Commission for Maritime Services (RTCM) SC-104 correction messages to the RBS which are subsequently relayed to the VBS following the J2735 standard.

Table 4 presents the different messages that are broadcast over the backhaul.

Table 4. Message Exchange over the Backhaul

Common Message Name	Message Standard	Sent by	Received
Signal Phase and Timing (SPaT)	J2735	RBS	CBS
MAP message (Geographic Intersection Description)	J2735	RBS	CBS
RTCM correction	RTCM SC-104	CBS	RBS
RBS System Diagnostics	JSON	RBS	CBS

Cloud service providers generally have advanced capabilities for high-volume data ingest, but many are tailored to Internet of Things (IoT) events that do not require real-time upload. For RCVW, however, messages are pushed to the cloud at the same frequency as they are transmitted to the VBS. The RCVW cloud provider offers two potential solutions for rapid incoming messages: Service Bus and Event Hubs. The Service Bus offering uses Advanced Message Queuing Protocol (AMQP) while Event Hubs use partitions like Apache Kafka. A Benchmark testing was conducted to determine the more appropriate solution for RCVW. See section 4.1 for more details

Cloud based functions are scalable serverless solutions for creating state-driven programs in the cloud. Each incoming SPaT message triggers a function execution for the specified HRI. When the HRI activation status transitions from inactive to active, or vice versa, the function creates a new HRI Event message. Depending on receipt of System Diagnostic messages, the CBS is capable of generating RBS Operational Status events for that HRI.

The RCVW CBS logs every MAP, SPaT, and RBS System Diagnostic message that it receives as well as every Event Message that it generates into a cloud-based data store. Event Messages are available in the data log for use in subscriber applications.

HRI Event messages including HRI Preemption Signal Status and RBS Operational Status, which are produced by the CBS are posted to an Event Streaming Endpoint to support the pushing of messages to a subscriber, for example a Traffic Management Center (TMC), listening for events. In Azure this is facilitated by the Azure Service Bus.

For every received MAP message, which includes the GNSS location of the HRI, the RCVW CBS initializes a proxy service that uses the Networked Transport of RTCM over IP (NTRIP) protocol to obtain GNSS corrections from a networked base station within the Constantly Operated Reference Station (CORS) network. These corrections are relayed back to RBS over the TCP/IP backhaul and ultimately broadcast over the RBS DCR. It is important to note that a failure of the proxy service or the backhaul network may cause the RCVW VBS to report an error state due to missing corrections.

Finally, as the system was being developed it was determined that HRI activation status information retrieval by a subscribed user of the cloud is best implemented through the existing

cloud provider's Application Programming Interface (API) management capabilities, allowing for simplified development for the subscriber applications.

The overall architecture of the RCVW did not experience changes as a result of the replacement of the DCRs from DSRC to C-V2X. The capabilities as well as the functionality of the system also remained unchanged.

Detail information on the Architecture and Design of the system can be accessed at the RCVW GitHub repository¹⁰.

3.3 RCVW Hardware Update

3.3.1 C-V2X Radios

The RCVW system was designed as a CV application with DSRC technology as the means of communication between the roadside and in-vehicle systems. On November of 2020, the FCC voted and approved to split the 75 megahertz of the DSRC radio spectrum and allocated the lower 45 megahertz of the band for unlicensed Wi-Fi use and the upper 30 megahertz for the use of transportation related communication technologies. The FCC requested that all DSRC licensees cease to operate in the lower 75-megahertz portion of the band by July 5, 2022, and transition all operations to the upper 30-megahertz. It also announced that it will request that transportation related operations in this sub band transition from DSRC to the emerging C-V2X technology. A specific date of when CV applications should fully phase out DSRC technology and adopt the C-V2X one has not yet been defined. However, in order to be ready by the time the FCC issues a final ruling on the matter, the Battelle team modified the RCVW architecture to incorporate C-V2X technology replacing the DSRC one on both the Roadside Unit (RSU) and the On-Board Unit (OBU).

3.3.2 FCC Request for Waiver of 5.9 GHz Band Rules to Permit Initial Deployment of C-V2X Technology

Prior to adoption of the final FCC C-V2X rules, any organization in need to broadcast within the specified C-V2X bandwidth was in need to request a waiver of the Commission's rules to allow the 5.905–5.925 GHz band to be used for C-V2X safety systems operating in the Intelligent Transportation System radio service. The Battelle team submitted the applicable request for waiver¹¹ on November 28, 2023. On April 18, 2024, the FCC granted Battelle and other companies a waiver request to deploy C-V2X technology within 5.895-5.925 GHz¹². This waiver allowed us to use our existing broadcast license to conduct C-V2X related tests.

3.3.3 C-V2X Vendor Outreach

Battelle submitted a Request for Proposal to vendors of C-V2X OBUs and RSUs to identify the best suitably technology for the project needs.

¹⁰ [RCVW/doc/RCVW_Architecture_and_Design.pdf at master · RCVW-PHASEIII/RCVW · GitHub](#)

¹¹ <https://www.fcc.gov/ecfs/document/112832256890/1>

¹² https://docs.fcc.gov/public/attachments/DA-24-363A1_Red.pdf

An analysis of the proposals was done, and the project selected the Danlaw AutoLink On-board units and Danlaw Routelink Roadside units.

Datasheets for the units can be found in Appendix A: Datasheets.

3.3.4 Computer Platforms

Both the VBS and RBS are supported by computer platforms which runs the RCVW application. The team researched the modifications and updates that the previous computer platforms have gone through and decided to update the system with units that provide greater processing power and speed. The team selected the following systems:

- RBS Computing Platform Units – Neosys POC-451 POC
- VBS Computing Platform Units – Neosys Nuvo-7204VTC

More details on these systems are provided on the Field Test and Evaluation Report [5]. Datasheets for the units can be found in Appendix A: Datasheets.

3.3.5 GNSS Devices

The RCVW updated the in-vehicle GNSS devices with a Ublox C102-F9R system. The F9R is equipped with an inertial measurement unit which is used to obtain the real time kinematic (RTK) solution required. Similar to the GNSS modules used in previous iterations, The device is capable of receiving multi-band multi-constellation signals and capable of reaching Real-time Kinematics (RTK) level of precision.

More details on these systems are provided on the Field Test and Evaluation Report [5].

Datasheets for the units can be found in Appendix A: Datasheets.

Table 5 shows the hardware evolution from RCVW II to RCVW-III

Table 5. RCVW Hardware Evolution

Hardware Function	RCVW-II	RCVW-III
On-Board Unit	Cohda MK5 OBU	Danlaw Autolink
Roadside Unit	Cohda MK5 RSU	Danlaw Routelink
In-Vehicle Computer Platform	Neosys-POC351VTC	Neosys Nuvo-72014VTC
Roadside Computer Platform	Neosys-POC351VTC	Neosys-POC451
In-vehicle GNSS device	Ublox-EVK-F9P-00	Ublox-C102-F9R

4 Testing and Test Results

4.1 Cloud Access Performance Testing and Investigation

The objective of this work was for the RCVW development team to find the most robust C/C++ Open-Source message broker client tools while simultaneously gain a perspective of the different protocols used to access the cloud. The first used an Apache Kafka client sending to and receiving from the Azure Event Hub, and the second used an AMQP protocol client sending to and receiving from Azure Service Bus.

4.1.1 Performance

The overall speed of delivery is considered to be the most important contributor to performance. Therefore, the tests tried to isolate individual steps in message processing so that each step could be measured more accurately. Thread synchronization was used to be able to 1) measure the true send-receive timing of each message, and 2) eliminate the use of wait “sleeps” that corrupt the understanding of the timings. Essentially, this keeps the machine busy queuing messages, instead of idling when waiting for each message to complete a round trip. The round-trip timing has at least 2 factors over which there is no direct control, but which can be isolated empirically. These are 1) network speed, and 2) processing time at the endpoint between receive and send back.

The performance testing showed that processing messages using the AMQP protocol against the Azure Service bus endpoint happens at the rate of 0.05 seconds per message. However, the test of Kafka against the Azure Event hub resulted in a factor of ~3 times longer (around 0.120 seconds compared to those of 0.05 seconds).

4.1.2 Client API Investigation

Different client APIs or Software Development Kit (SDK) were studied to learn which might be best for current and future V2X-Hub prototype development. For the AMQP protocol, 3 different API2s were investigated:

1. RabbitMQ, formerly called rabbitmq-c
2. AMQP-CPP, a C++ RabbitMQ client (CopernicaMarketingSoftware, 2024)
3. Qpid-Proton and Apache project (Apache Software Foundation/Qpid Proton, 2024)

Qpid-Proton proved to be the best choice for the project needs. Implementing the cloud communications making use of the advantages that Qpid-Proton provides has significant benefits:

- The code communicating with the endpoint is completely independent of the client it is serving.
- Many systems do not need “logic” within them for allowing access to the endpoints. That capability is handed completely over to a black box with simple interfaces.
- The communication to the endpoints can be offloaded to different processes in a simple manner as the needs arise.

- The type of programming patterns fostered with distributed systems create extremely robust, flexible, and efficient systems

Detail information on this performance test can be found at the RCVW GitHub repository¹³

4.2 Cloud-based Subsystem Bench Test

The implementation of the CBS requirements to the RCVW system provided the following capabilities:

- The RCVW can now serve as a tool for an authorized user to remotely monitor and/or receive information on the activation status of the HRI warning devices based on the data received from the HRI controller and the RBS.
- An authorized user can access the RCVW CBS database and query historical train presence at equipped HRIs.
- An authorized user can query the CBS to gather Identification Number (ID)s of HRIs in a pre-determined radius.
- The RCVW can serve as a tool to provide static HRI information such as the one provided in the FRA Grade Crossing Inventory System.¹⁴

Figure 3 shows the CBS Database views that enables the designed capabilities:

¹³ <https://github.com/RCVW-PHASEIII/RCVW/blob/master/doc/Cloud%20Access%20Performance%20Testing%20Document.pdf>

¹⁴ <https://safetydata.fra.dot.gov/Authentication/AuthSelection?AppID=GCIS>

These capabilities were tested during the CBS bench test evaluation (BTE) event. The BTE served to confirm the system requirements specified for the RCVW CBS. It provides the methods followed to verify and demonstrate that the CBS has the functionality and performance capabilities necessary to deliver the features described in the RCVW Concept of Operations document [1] and that the system meets the functional and performance requirements described in the RCVW System Requirements Specification document [2].

- Test execution of the RBS and CBS systems was conducted primarily by two staff. One individual handled testing the software, which primarily consisted of using a simple navigation application alongside the RBS/CBS to test a predetermined list of functionalities. A second individual oversaw software development and bug management. As the first individual would find issues with the system, they would create software bug tickets describing the issue and send them to the second individual, who would then go through the necessary processes to fix and deploy the software back to the system. An API that provides a web interface to simulate and visualize a vehicle approaching an HRI was used on several test cases. This interface also enabled the operator to select a route for the vehicle to traverse.

Table 6 presents a summary of the test cases conducted:

Table 6. Summary of RCVW BT&E Test Cases

Test ID	Test Case Title
1.0	RCVW Architecture and Message Verification
1.1	Navigation System HRI Notification - Functioning Connection between CBS and RBS - HRI Warning Devices Activated
1.2	Navigation System HRI Notification - Functioning Connection between CBS and RBS - HRI Warning Devices Deactivated
1.3a	Navigation System HRI Notification - CBS and RBS Communication Link Failure Warning
1.3b	Navigation System HRI Notification - CBS and Navigation System Communication Link Failure Warning
1.4	Navigation System HRI Notification - HRI Not Equipped with RBS – may or may not have active protective devices
1.5	Navigation System HRI Attributes - Low Grade Crossing En Route

Test ID	Test Case Title
1.6	Simulated Traffic Management Center HRI Activation Status Change Notification
1.7	Simulated Traffic Management Center – CBS and TMC Communication Link Failure

The BTE required a follow-up regression test activity to validate RBS specific requirements. See Table 7.

Table 7. Regression Testing Requirement Verification

Req. ID	Requirement Title
RBS-017	The RBS shall broadcast its own system diagnostics information to the CBS at least once per second during operation.
CBS-012	The CBS shall be capable of storing RBS Diagnostic information for HRI's linked to an RBS.
CBS-015	The RBS diagnostic information shall indicate the operational status of the RBS components and overall RBS software modules

4.2.1 Summary of RCVW CBS Performance

The primary focus of this round of bench testing was to verify both data ingest and data retrieval at the CBS. Messages sourced from the RBS were received by the Azure Service Bus running inside the cloud. This was confirmed by ensuring that the data sent from the RBS was stored inside the correct Service Bus subscription queue. Likewise, the test navigation application was utilized to request crossing data from the CBS using a secured RESTful API. Confirmation that the returned data is equivalent to that which is stored in CBS was accomplished by database inspection.

Test cases that failed during the first round of bench testing were marked and tickets were created using the Jira software. This allowed for quick and concise communication between testers as well as an organized system for tracking the current progress of each issue. Tickets that were resolved within a timely manner were then retested for confirmation. The failures listed below were retested and confirmed prior or during the demonstration event (See section 5) or during the Field Test Event (See section 4.3).

Detail information on the bench test can be found in the *RCVW Cloud Based Subsystem Bench Test & Evaluation Report* document [4] within the RCVW GitHub repository.¹⁵

Table 8. Summary of RCVW BT&E Detected Issues and Resolution Status

#	Jira ticket #	Issue	Resolution Status	Explanation
1	TMX-133	Azure Service Bus topic names for plugin static/config/error channels should match the plugin name from the manifest	[Deferred]	Only may become a concern if more than one RBS exists
2	TMX-134	Azure container app not deployed correctly	[Fixed]	Ran local instances in part during bench test. Cloud deployment reworked using Docker containers for better integration.
3	TMX-135	RSU Immediate Forward plugin not sending messages to radio	[Fixed]	The plugin operation was corrected so the messages received were also forwarded to the RSU
4	TMX-136	SPAT messages are not sending fast enough and there are multiple gaps of more than a tenth of a second.	[Fixed]	Seems to have been from the logging API. Made some fixes, but ultimately resolved by only using the root logger. Fixed prior to Demonstration event.
5	TMX-137	No RTCM messages being sent	[Fixed]	Need an Azure container app that maintains a connection to CORS on behalf of the RBS. Fixed prior to Field Test Regression Event
6	TMX-138	When running the RBS, messages are being sent regarding a full AMQP queue on the sender	[Fixed]	Reworked the QPID Proton broker and retested over a day period.
7	TMX-139	IEEE-1570 Message Type 4904 not being logged in the CBS, in violation of [CBS-007]	[Fixed]	There was no specific client need for this requirement, although it seems like it could be useful. Fixed prior to Field Test Regression Test
8	TMX-142	Bench test 1.3a is a test about using the navigation while the RBS is disconnected from the Battelle network. The app is supposed to display that the RBS is not connected and should display 'unknowns' for HRI values. This did not happen	[Fixed]	This is a problem in the navigation app, which is not being maintained. Therefore, it will not be resolved at the time being. Fixed prior to Demonstration event.

¹⁵ [RCVW/doc/RCVW_Cloud Based System Demo Report_Final.pdf at master · RCVW-PHASEIII/RCVW · GitHub](#)

#	Jira ticket #	Issue	Resolution Status	Explanation
9	TMX-143	Bench test 1.5 states that the app should have a feature that allows the user to decide to receive notifications related to a "low-grade crossing" attribute of an HRI. There is a single data element labeled "Low Grade" in the "HRI" panel, however this data element is always either empty or "unknown".	[Deferred]	This is a problem in the navigation app, which is not being maintained. Therefore, it will not be resolved at the time being. Overcome by events. A different application was used during the demonstration event.
10	TMX-144	The diagnostic messages from the HRI Status plugin and the RSU Immediate Forward plugin were not being sent at least one time per second.	[Fixed]	Added more frequent diagnostics.
11	TMX-145	SPAT message topic missing from Azure Service Bus	[Fixed]	The topic was somehow removed accidentally during testing. It has been restored.
12	TMX-146	TMC scenario not deployed to Azure	[Deferred]	The TMC scenario testing from bench tests 1.6 and 1.7 was done on a local instance and needs to be added to the cloud. This scenario was implemented and tested during the demonstration event.
13	TMX-147	CBS generated HRI Event Messages are missing	[Open]	The navigation app scenario testing from bench tests 1.1 through 1.5 was done using the direct DB update feature of the navigation app.
14	TMX-148	Compare diagnostic message content against requirements	[Open]	Need to verify the diagnostic message contents against the RCVW requirements
15	TMX-149	Verify methods of cloud storage and cloud service analytics	[Open]	Need to confirm the specific Azure guarantees against the RCVW requirements
16	TMX-150	Compare HRI data against the FRA data dictionary	[Deferred]	Broken link on FRA web site made this impossible.
17	TMX-151	Integrate IEEE-1570 device into RBS test equipment	[Open]	Ensure that flipping the switch modifies the SPAT message on the RBS and in the CBS and generates an HRI event.
18	TMX-152	Add field for the HRI data endpoint URL. [CBS-017]	[Open]	The CBS generated HRI events are sent to an Azure Service Bus topic, but those endpoint details

#	Jira ticket #	Issue	Resolution Status	Explanation
				are currently hard-coded and not available in the API
19	TMX-153	Clarify requirement [CBS-017]	[Deferred]	As written, the requirement looks too much like a web URL, when in reality the information is an Azure Service Bus topic

4.3 C-V2X Field Test

The RCVW system went through a detailed Field Test as part of the Phase II of this project back in October of 2020. During Phase-II testing, it was demonstrated that the RCVW system possesses the functionality and performance necessary to deliver the functions and benefits proposed for the system in an operational setting and verified that the RCVW system meets the requirements documented in the System Requirements Specification. Details and results of this test event are described in the RCVW Phase II Field Test and Evaluation Report. [7]

For this phase of the project a field test and evaluation (FT&E) event was performed to verify and demonstrate that the C-V2X radios and the updated hardware allow the RCWV to achieve the necessary performance to deliver the features described in the RCVW ConOps document [1].

The FT&E was conducted at Battelle’s West Jefferson facilities in West Jefferson, OH, on July 25–26, 2024, and on December 12-13, 2024. Prior to test execution, Battelle staff installed an RBS and RSU at a dedicated 0.4-mile test road located in Battelle’s West Jefferson campus. The RCVW RBS was housed in a traffic cabinet located at the end of the test road (outlined by the red-colored rectangle in Figure 4. An Institute of Electrical and Electronics Engineers (IEEE)-1570 compatible communication system was used to generate the train-detection preemption messages. An ethernet switch was used to connect the radio and RBS over a dedicated internet connection behind Battelle’s firewall. This configuration allowed remote connectivity to the RCVW system by the Battelle team which was used to observe system performance and troubleshoot any issues remotely.



Figure 4. RCVW RBS

Battelle mounted the C-V2X RSU radio on a metal post located adjacent to the cabinet (outlined by the yellow-colored rectangle in Figure 5). An RTK base station antenna was positioned on a corner on top of the cabinet (outlined by the blue-colored rectangle in Figure 5). Four 110 Vac power outlets within the cabinet were used to provide power to the RBS, RSU (via a Power Over Ethernet (POE) device), switch and IEEE-1570 device.



Figure 5. RCVW RSU

A 2009 Mazda 3 vehicle was equipped and used for all the field test cases. See Figure 6.



Figure 6. Test Vehicle

The test was conducted by three Battelle team members. One Battelle team member monitored the algorithm operation, remotely accessing the Vehicle's RCVW systems when needed for any technical aspects and resolution of issues and assisted when features needed to be turned off. A second Battelle team member was responsible for collecting the start and end times for each test run start and ending, notifying the CV driver of the test to be performed, taking notes of any irregularities communicated by the driver and support in reconciliation activities. A third team member was dedicated to operating the vehicle, which involved driving the vehicle straight along the approach, reaching and maintaining the designated vehicle speed. A speed of 35 mph was selected after several test runs were conducted and the driver observed that higher speeds could pose a safety concern for vehicle control and deceleration (due to the test track geometry).

4.3.1 Approach to Scenario-Based Field Testing

A total of 14 scenario-based test cases were performed; 11 consisted of 3 runs approaching at 35 mph, 1 consisted of the vehicle travelling outside of the test track to measure radio signal coverage, 1 test case had the vehicle approach, stop and remain within the HRI hazard zone and 1 stability test was performed in three stages which were aimed at assessing the level of nuisance alerting observed with the vehicle stopped at various locations near/within/beyond the HRI Hazard Zone. Of the 14 test cases, 5 were fault verification test cases.

Table 9 summarizes the conditions and inputs for each of the scenario-based test cases executed in FT&E.

Table 9. Field Test Cases

Test Case Number	CV Location at Preemption Signal Detection	CV Speed at Approach Entry	RCVW State
1.1	N/A No preemption	35 mph	Normal
1.2	Start of Approach	35 mph	Normal
1.3	$\frac{3}{4}$ point	35 mph	Normal
1.4	Midpoint	35 mph	Normal
1.5	Near HRI Hazard Zone	35 mph	Normal
1.6	Start of Approach, then preemption is removed at Midpoint	35 mph	Normal
2.1	Stopped on Tracks (preemption will deactivate during the test)	N/A	Normal
3.1	Start of Approach	35 mph	Fault – C-V2X
3.2	N/A	35 mph	Fault – C-V2X
3.3	Start of Approach	35 mph	Fault – GPS
3.4	Start of Approach	35 mph	Fault – MAP
3.5	Start of Approach	35 mph	Fault – IEEE-1570
4.1	N/A No Preemption required	N/A	Normal
4.2	N/A No Preemption Required	N/A	Normal

Several issues and anomalies were identified during the Field Test activities. The development team analyzed each issue and implemented several software code corrections to ensure that all expected requirements and capabilities are met. A regression Field Test activity was performed on December 12 and 13, 2024 at the West Jeff location where specific test cases were selected and can be found in Table 10.

Table 10. Regression Test Cases

Test Case No.	Test Case Title
1.2	CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions

Test Case No.	Test Case Title
1.3	CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions
2.1	CV Stopped in HRI Hazard Zone with Preemption Active/Not active
3.1	Loss of C-V2X after CV has received an RCVW
3.3	Loss of RTK Fix (GPS)
3.4	Loss of MAP
3.5	Loss of IEEE 1570 Heartbeat
4.2	GNSS Stability and Accuracy (Observation)

These test cases were selected to ensure that the following is verified:

- Confirm the reception and application of RTCM corrections in the VBS.
- RCVW system being capable of producing distinct alerts when a vehicle is within the HRI and preemption signal is On/Off.
- Stability of the GNSS module when RTK corrections have been applied.

4.3.2 Summary of RCVW Performance

After the FT&E and regression FT&E was conducted these are the results.

- 1) In summary all requirements were confirmed to have been met with the exceptions presented in Table 11. Requirements not met¹⁶

Table 11. Requirements not met

Requirement ID	Requirement Text
VBS-022	The VBS shall be capable of receiving messages from the RBS within 50ms
RBS-004	The RBS shall broadcast the HRI Activation Status message to both the VBS and CBS 10 times per second during operation.

¹⁶ For details on the verification activities see RCVW C-V2X Field Test & Evaluation Report within the RCVW GitHub repository [RCVW/doc/RCVW-III Field Test and Evaluation Report_Final.pdf at master · RCVW-PHASEIII/RCVW · GitHub](#)

Requirement ID	Requirement Text
VBS-012a	The VBS shall continually issue unique HRI Hazard informs to the vehicle operator while the HRI safety devices are not active and the vehicle is stopped within the HRI Hazard Zone.
RBS-005a	When the RBS is connected to an IEEE-1570 compatible device and the Heartbeat signal is lost, the RBS shall cease sending HRI State Messages.

- 2) The RCVW RBS is capable of receiving RTCM corrections from its corresponding CBS. These corrections are analyzed and broadcasted over the DCR to the in-vehicle subsystem for it to apply to its position solution.
- 3) Test cases 3.1 to 3.5 were designed to verify the correct issuing of system faults when critical data was not received or received at a different rate than what is required. When the necessary fault message was generated, the system immediately cleared this fault message. A few seconds later, the error was reissued and once again, cleared by the system. This continued until the fault was extinguished. Further investigation is required.
- 4) The RCVW system is configured to issue a fault when the following issues with the GNSS signal are detected:
 - a) Loss of GNSS signal
 - b) Loss of RTK fix
 - c) Position data reception at a rate less than 10hz

During testing, the VBS would issue a Loss of RTK fix fault message at random times. After an investigation it was discovered that the GNSS device would lose configuration during these situations resulting in the RCVW correctly issuing a fault due to the loss of RTK fix. In order to solve this issue, the GNSS device needed to be reprogrammed. Further investigation is required.

Overall, the FTE demonstrated that the update in the hardware and the inclusion of the CBS to the system allows for the RCWV to achieve the necessary performance to deliver the features the system was designed to fulfil.

More information on the test procedures and results can be found in the RCVW C-V2X Field Test & Evaluation Report within the RCVW GitHub repository ¹⁷

¹⁷ [RCVW/doc/RCVW-III Field Test and Evaluation Report_Final.pdf at master · RCVW-PHASEIII/RCVW · GitHub](#)

5 CBS Short-term Demonstration

As part of this project, a short demonstration of the Cloud-based subsystem was designed and deployed from 09/11/2024 to 12/06/2024 in the city of Jacksonville, FL. The goal was to deploy four RCVW cloud-based subsystems in a region where vehicle traffic was often disrupted by HRIs that were closed due to the presence of a train and take advantage of the cloud-based capabilities to inform remote users of these blockages.

5.1 Participation

The Battelle team reached out to the Florida Department of Transportation (FDOT) District Two with a proposal to be part of this demonstration. The city of Jacksonville was selected as the demonstration site and FDOT provided coordination and selection of the HRI sites to be equipped as well as the installation and retrieval of data. Trimble provided the Battelle team with a developer's license and an API key for their CoPilot tool. This enabled the use of Trimble maps to create a Graphical User Interface (GUI) that served as the user interface for the CBS.

5.2 Demonstration Scope

The RCVW CBS was designed to provide HRI information to subscribed users over the internet. It receives and stores timely HRI information (including HRI warning device activations) which can then be pushed or retrieved by authorized users. Being able to access information about an HRI blockage allows drivers to choose alternative routes or delay the start of a trip until tracks are clear, minimizing delays and congestion. In the case of an emergency, knowing whether a crossing is blocked helps emergency responders plan their approach and avoid delays.

For this demonstration, the CBS provides the following information:

- Alert cloud subscribers of the activation/deactivation of the monitored HRI protective devices
- Inform cloud subscribers of the communication health between the RCVW unit monitoring the HRI and the CBS
- Provide cloud subscribers the following Static HRI-related information
 - Crossing ID
 - City
 - County
 - Street
 - Railroad Organization
 - Emergency Contact Number
 - Railroad Contact Number
 - Latitude and Longitude

- Mile Marker
- Distance to Highway
- Speed limit
- Crossing Angle
- Crossing Position
- Gate Presence
- Flashing Light Presence

The Battelle team used the provided Trimble API to develop a GUI intended to simulate a TMC. A TMC is a central hub from which traffic is monitored and managed in real-time. The primary purpose of a TMC is to optimize the flow of traffic, improve safety, and minimize the effects of incidents that occur on the road.

By having access to timely HRI-related information, a TMC would be capable of the following:

1. **Safety Monitoring:** Monitoring train movements at HRIs allows the TMC to ensure safety for both vehicles and pedestrians. By detecting when a train is approaching or occupying a crossing, the TMC can provide timely warnings to drivers and pedestrians, preventing accidents.
2. **Traffic Flow Optimization:** Knowing when an HRI is potentially blocked due to a train helps manage traffic flow. The TMC can adjust traffic signal timings, reroute vehicles, or provide real-time information to drivers, minimizing congestion and delays.
3. **Emergency Response Coordination:** In case of emergencies (e.g., stalled trains, derailments, or accidents), the TMC needs accurate information about train locations. This enables efficient coordination with emergency services and ensures a swift response.
4. **Infrastructure Maintenance:** Monitoring HRIs helps track wear and tear on the infrastructure. Regular maintenance can be scheduled based on usage patterns, reducing downtime and ensuring safe crossings.

The simulated TMC was represented by a GUI presenting a map view of the city of Jacksonville, FL. The map indicates the location of HRIs in the area and in particular four locations equipped with the RCVW CBS.

An authorized user of the TMC would click on one of the sites to access the HRI-related information. When the CBS receives data from the RCVW RBS indicating that the HRI is active, the GUI issues an audible alert and changes the icon of the location to indicate as such. When the active warning devices in the HRI deactivate, the icon returns to its original shape and color.

When a monitored site loses connectivity with the CBS, it changes the icon indicating a problem is detected. The user can then click on the site and information related to the situation is presented. All the data related to the HRI (activations, Comm loss, deactivations, etc.) is stored in the CBS for future analysis and retrieval.

Four locations in the City of Jacksonville were selected that matched the following characteristics:

- Location consists of an HRI that experiences 10 or more activations in a period of 24 hours.
- Selected HRIs are located within 200 feet of a signalized intersection.
- Traffic signal controllers at the signalized intersection are receive a signal preemption from the HRI controller when a train is approaching the crossing.
- Preempted intersection experience often experiences traffic disruptions due to HRI closings

The RCVW is designed to monitor the status of the preemption signal sent by the HRI controller and generate and broadcast changes in SPaT whenever there is a change in preemption status. Typically, the HRI preemption signal is interconnected with an HRI equipped with active warning devices is located within 250 feet of a signalized intersection. When a train approaches the HRI, the HRI controller preempts the traffic signal controller thereby initiating a sequence of steps to clear any potential highway vehicle traffic stopped within the collision zone of the HRI.

5.3 Planning

Four locations in the city of Jacksonville, FL. matched the needs of the demonstration were identified See Table 12:

Table 12. Crossing Information

Crossing ID	City	Street	Latitude/Longitude
621213F	Jacksonville, FL	San Juan, Ave	30.282196/-81.720559
621221S	Jacksonville, FL	St. Johns Ave	30.2885320/-81.7206070
271824W	Jacksonville, FL	Sunbeam Rd	30.2060770/-81.5787700
621275X	Jacksonville, FL	S Edgewood Ave	30.3269015/-81.7308415

Based on crossing closure data provided by FDOT each crossing included in the demonstration experiences an average of ten crossing warning device activations per day due to train presence.

The RCVW RBS was designed to receive preemption signals from the HRI controller to detect a train approaching. The RBS follows the common track circuit design where a constant 24 Volts Direct Current (VDC) indicates train not present while a drop in the voltage indicates train present.

For this demonstration, the Battelle team selected sites where an HRI provided a preemption signal to a traffic signal controller nearby when a train is approaching. When the preemption signal is removed, the Traffic Signal Controller closes a load switch and completes a 120 Volts Alternate Current (VAC) circuit that commences the traffic signal sequence to clear traffic from the HRI. For this demo, the RCVW system used this 120 VAC signal from the load switch to open a normally closed relay which provides the required 12 VDC signal to the DB-15 digital

I/O (DIO) interface on the RBS (via a power supply). This indicates the presence of a train to the RBS. (Figure 7)

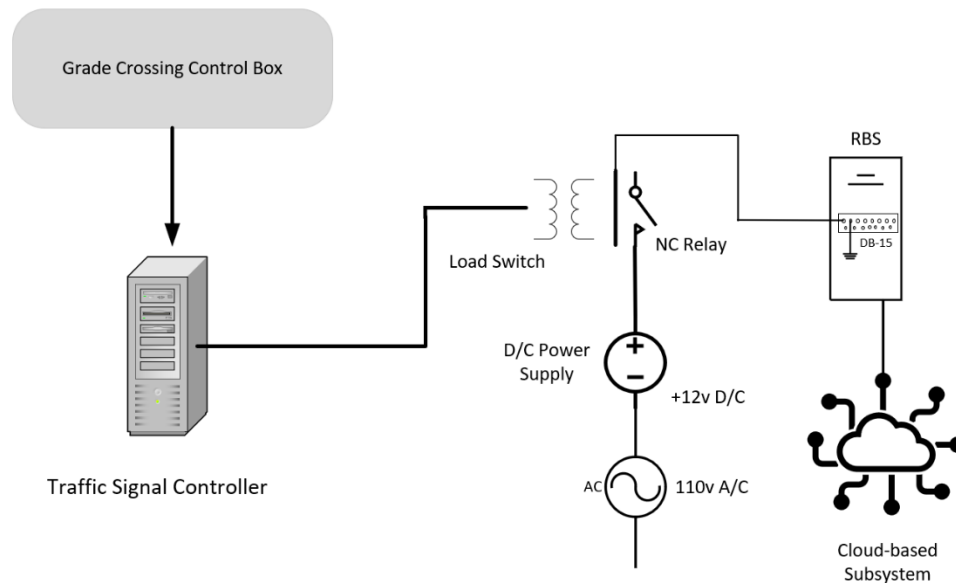


Figure 7. RBS Interface with Traffic Signal Controller

The RBS was connected to the CBS via a dedicated wired internet connection provided by FDOT. However, at the St. Johns location, an internet hotspot interfacing via the mobile network was used. See the Data Analysis section for details. The CBS had each of these locations pre-programmed in the cloud with its corresponding HRI ID, SAE J2735 MAP file, Latitude and Longitude ready to receive HRI-related information. These sites were pre-loaded as well with HRI-related attributes from the FRA database.

Figure 8 shows a graphic representation of the RCVW configuration for this demonstration.

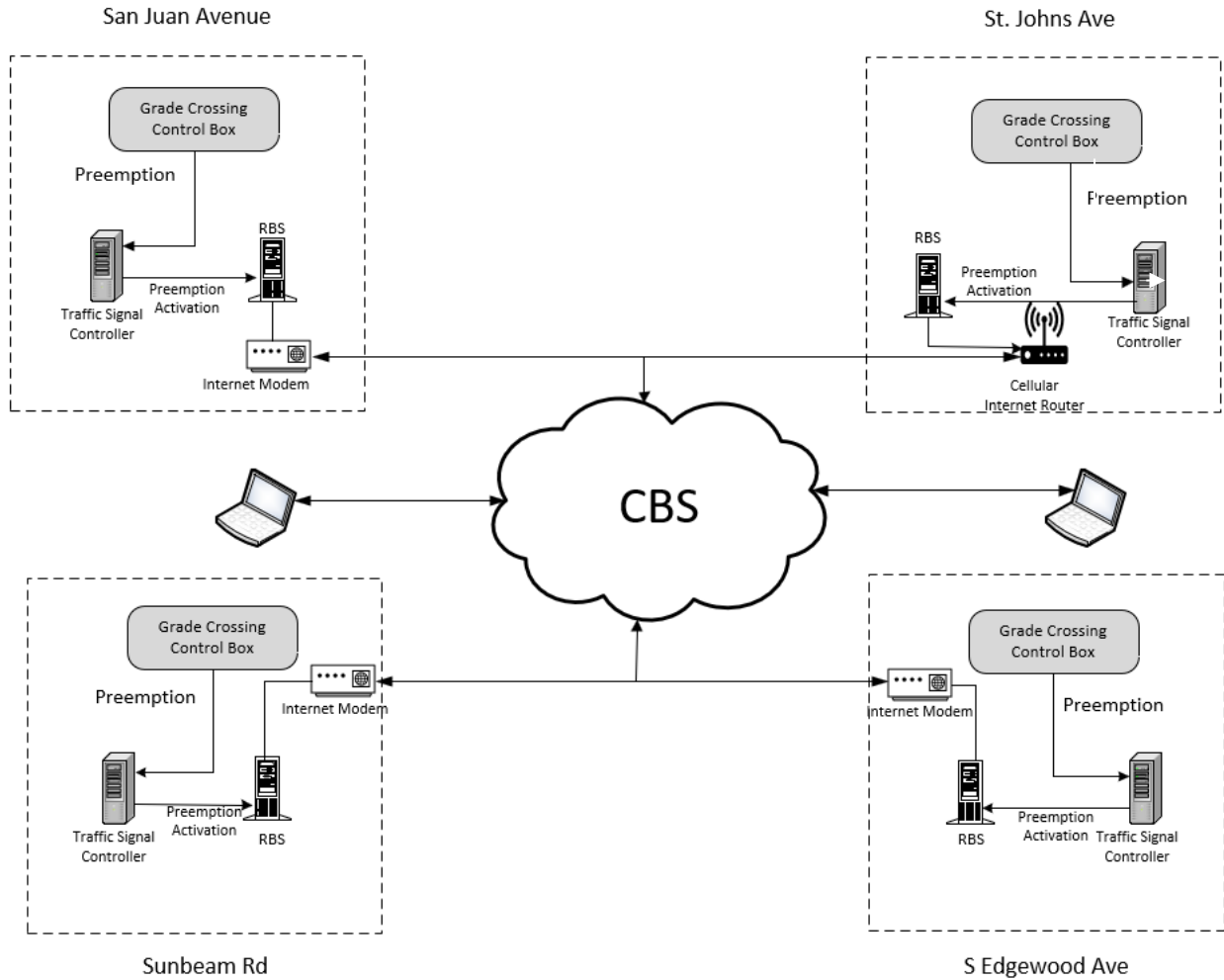


Figure 8. Jacksonville, FL RCVW configuration

5.4 Demonstration Tools

The GUI used for this demonstration was designed to display a map of the area along with icons representing HRIs and information for each of the locations. This map allows the user to zoom in and out of the location, move and center the map to a particular point as well as provide a graphic and audio alert when a monitored HRI detected the arrival of a train. The goal for this tool was to serve as a simulated TMC monitoring HRIs on real time (Figure 9). This TMC acted as an authorized subscriber of the CBS and was monitoring the four equipped location. Users with the proper credentials were able to access the service remotely and were presented with a graphical representation of the area where the monitored HRIs are located as well as real time status of the four equipped locations.

In detail, the GUI enabled the following capabilities:

Table 13. GUI Capabilities

Capability	Details
Graphic representation of the four RCVW equipped and several non RCVW equipped HRIs	All RCVW-equipped HRIs were represented on a map by specific distinguishable icons. All non-RCVW-equipped HRIs were represented on a map by specific distinguishable icons
Graphic representation of the current communications health of the HRI monitored equipment	The color of the monitored HRI icons changed depending on the communications link between the CBS and the HRI.
Graphic representation of the current activation status of the active warning devices of the monitored HRIs	The color and icon of the monitored HRI icons changed depending on the status of the warning devices
Audible alerts	When the CBS detected the HRI status to be active, a short audio alert was issued.
Information of all HRIs	HRI ID HRI Location Activation Status Type of warning devices present Any other information that is presented in the FRA data dictionary can be accessed by the user

In addition, the GUI allowed a user to manually simulate activations and loss of communications on specific locations to observe the CBS functionality on demand.

Table 14 describes the different icons and the information they provide

Table 14. GUI Icons

Icon	Details
	Yellow icons indicate HRIs in the area. When the user selects one of these icons a pop-up window with HRI information appears.
	Blue icons indicate HRIs that are being monitored by the CBS.
	Red Icons indicate the HRI warning devices are active.
	Icons with an overlaying "X" indicate a loss of connectivity between the CBS and the HRI location.

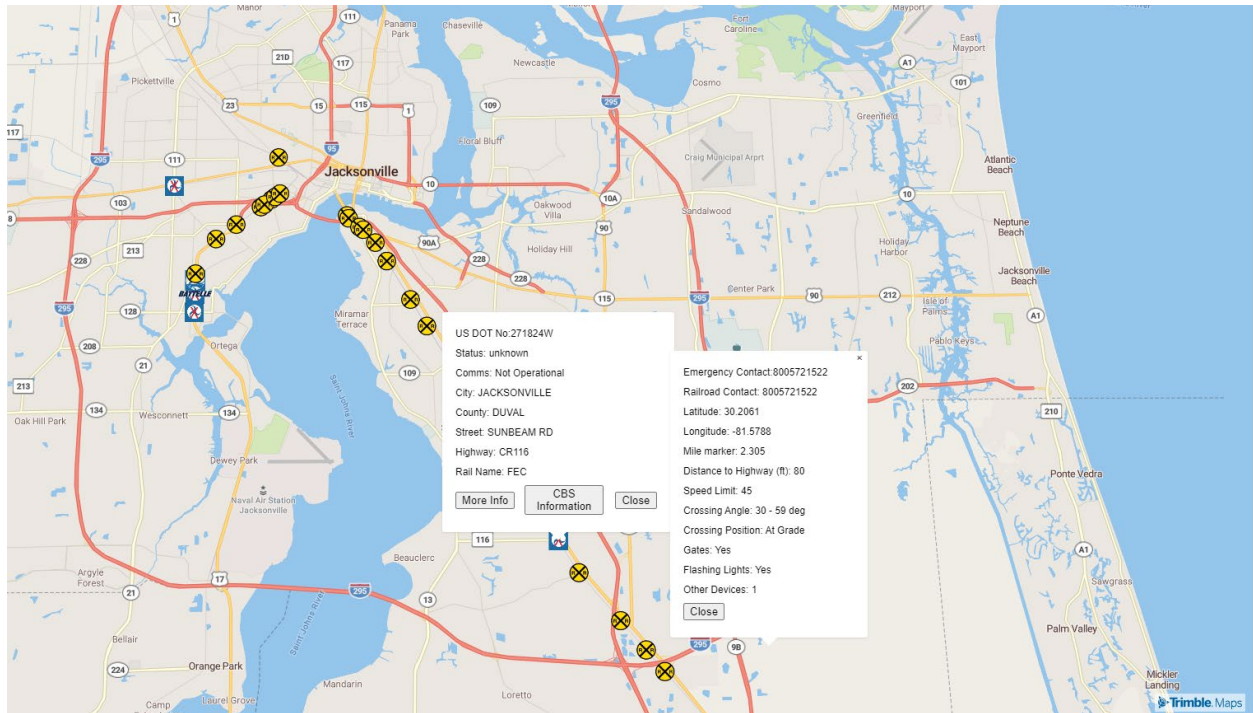


Figure 9. Simulated Traffic Management Center

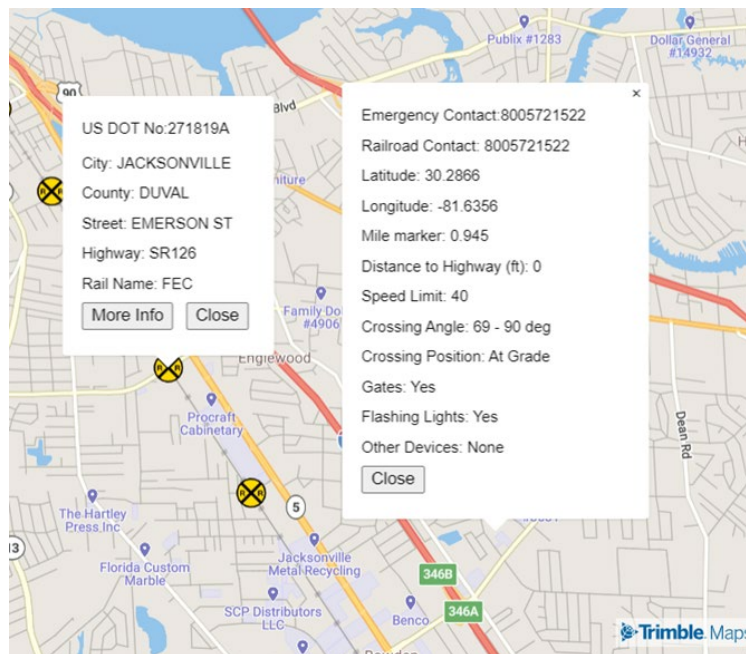


Figure 10. HRI Information

The yellow icons indicate HRIs in the area. When the user select one of these icons a pop-up is displayed.

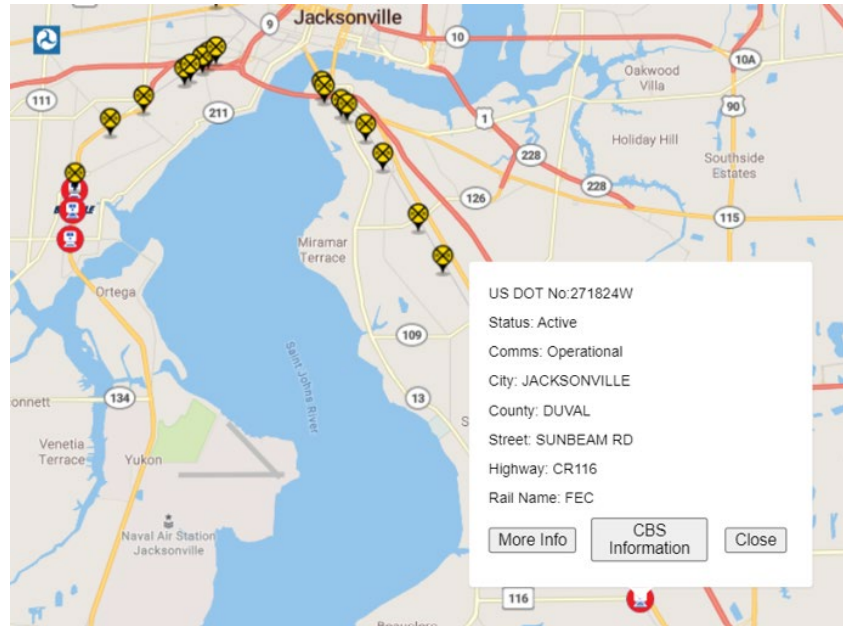


Figure 11. CBS Equipped HRIs

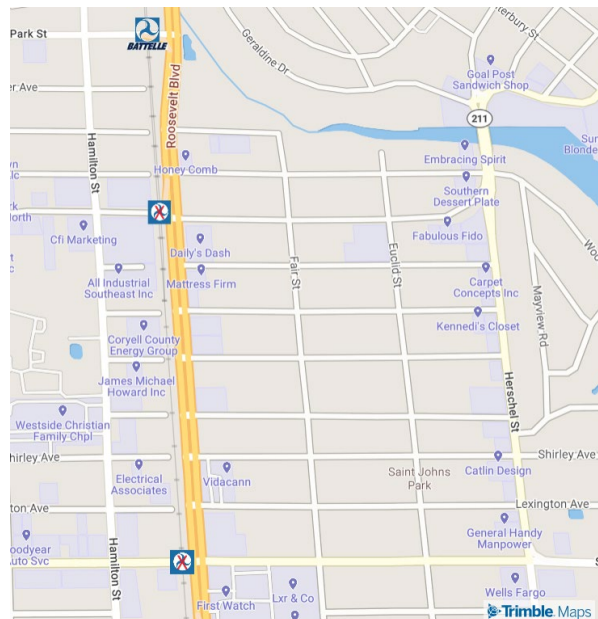


Figure 12. Loss of Comms to Equipped HRIs

5.5 Traffic Camera Video Display

In order to verify that train presence at the monitored locations were being correctly registered by the CBS, FDOT provided access to a traffic camera located at the St. Johns location. This tool allowed the Battelle team to visually verify when the web service indicated a train was approaching. Figure 13 presents a train arriving to the St. John's HRI. The active warning devices are activated, and the web service has indicated (by changing the icon to a red color and adding a shape of an incoming train) that an incoming train has been detected.

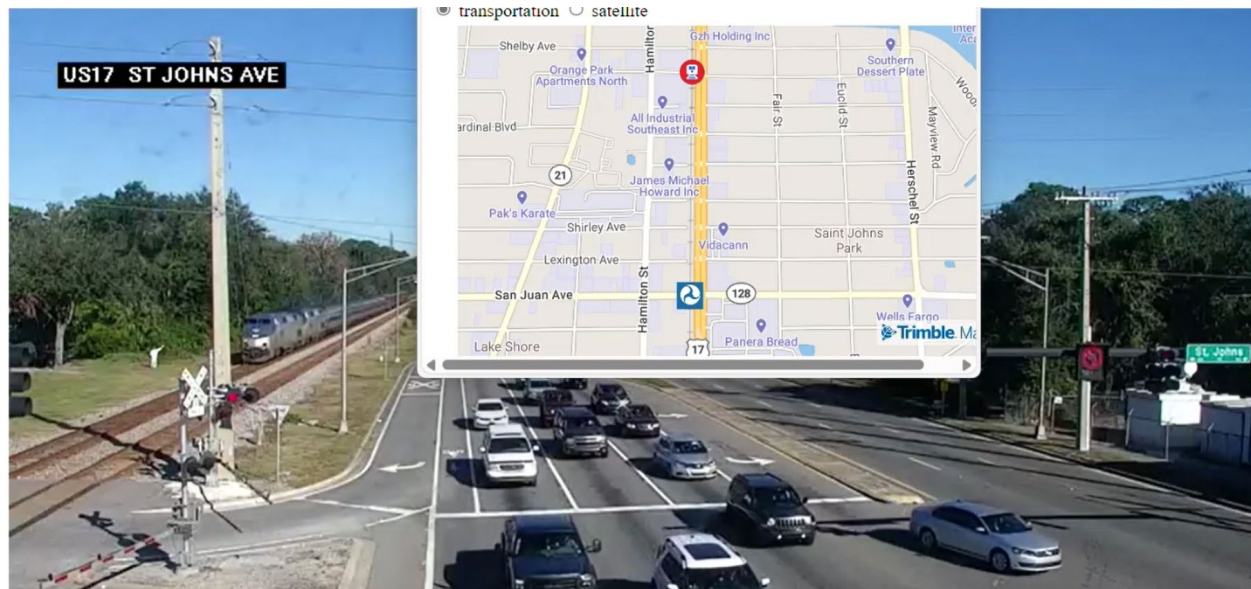


Figure 13. Train approaching St. Johns location

5.6 Data Analysis

During the time the CBS was active in the city of Jacksonville, it collected the following information from the four equipped locations:

- Date and time of change in SPaT indicating the change in the pre-emption signal
- Communications link status
- All Communication logs between RBS and CBS

The Communication logs between the RBS and the CBS were collected on a 24/7 basis. These logs provide information regarding each HRI location such as:

- Time a status message was created in the RBS
- Time a status message was received at the RBS
- Indication of pre-emption signal change

This data was used to produce an analysis of the latency encompassing the time from the detection in the loss of the pre-emption signal through the time it is registered and processed in the CBS.

5.7 Results

Overall, all four systems prove to comply with the CBS requirement of 0.5 seconds (500 milliseconds) for message receipt and 5 seconds (5000 milliseconds) for the entire process, including CBS ingest performance. An analysis was also done to confirm the CBS activations matched train activation data provided by FDOT. Data analysis for the period of 10/18/2024 through 11/14/2024 shows that the CBS correctly matched 1746/1792 activations/deactivations (97.4%) during days when the systems were actively tracking crossing activity.

5.8 Challenges

There were a few challenges the Battelle team faced during this demonstration of the RCVW CBS.

5.9 Environmental Issues

During the demonstration period, the CBS experienced periodic communications issues with the equipment installed at the four traffic signal cabinets. One possible cause of this was instability within the City of Jacksonville intranet routing. A proposed solution to mitigate this problem in one location, was to install a mobile network hotspot to allow a wireless internet connection of the RBS computer. Extreme temperatures inside the traffic signal cabinet in Florida were affecting the functionality of the mobile hotspot.

A rugged Long-Term Evolution (LTE) router was installed that could withstand the temperatures inside the cabinet. There was no further temperature related connection issues.

5.10 Connection Issues

Following periods of network disruption, the RBS failed to automatically reconnect to the CBS to continue delivery of the messages. After an investigation and to remedy this situation the process on each RBS was periodically restarted every 30 minutes (at :15 and :45 past the hour) in order to ensure new connections were established. The restarts were relatively fast, thus would generally not affect data analysis during normal operating periods. However, the data analysis had to filter out clear indicators that a restart was occurring.

Late in the demonstration cycle, new code was produced to build a sender with its own connection object, which makes it possible for automatic reconnection to occur. However, this was not verified during the demonstration period due to time constraints.

5.11 Time Synchronization

The current CBS configuration and interface does not allow for direct time synchronization with the rest of the traffic management devices. This resulted in the inability to perform a complete analysis of the performance of the system in comparison to the tools currently used.

To achieve this, the CBS needs to be fully integrated into a deploying organization's network and allowed to be managed from within. However, even though the CBS was not considered as part of the network architecture, the system was proven to be dependable and accurate.

Detail information can be found in the RCVW Cloud-based System Demonstration Report within the RCVW GitHub repository ¹⁸

¹⁸ <https://github.com/RCVW-PHASEIII/RCVW/tree/master/doc>

6 Conclusion

The inclusion of the Cloud-based subsystem to the RCVW expands its capabilities by allowing the broadcast of active HRI warning devices activation status over the internet to any authorized subscriber.

Results from the data collected during the demonstration of the CBS indicates that some train detections were missed and the integrity and accuracy of the data used to compare the train detections was not performed. To obtain an accurate comparison, detections would need to be evaluated in a controlled environment where the integrity of the data is verifiable.

Based on the results of a limited Field Test and Evaluation event performed as part of this project, the update of the CV communication radios to a C-V2X technology did not have a noticeable effect on the functionality of the system and the performance was on par with previous experiences.

This project provided the necessary data to confirm that the RCVW CBS:

- can inform remote users of the activation status of HRI warning devices..
- is capable of interfacing in multiple ways with traditional traffic management devices in the field to obtain the activation status of HRI warning devices.
- is capable of storing HRI active warning device activation state changes events as well as static HRI data which can be retrieved for analysis
- does not require a direct connection to Railroad crossing control equipment to obtain HRI active warning devices activation status.
- does not require the installation of additional train sensing equipment
- seamlessly interface with existing DOT and municipality traffic equipment with no disruption to the traffic management activities.

The system is still considered a prototype requires further refining to become a reference implementation.

The next logical step is to promote the benefits of or RCVW and generate industry acceptance by conducting an outreach campaign to different stakeholders These include local and State agencies, vehicle manufacturers and railway companies. With their support, model deployments and pilot installations across the United States can be performed to evaluate the efficacy of the system in real life scenarios.

Referenced Documents

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2. Sanchez-Badillo, A., Baumgardner, G (2023). Rail Crossing Violation Warning. System Requirements Specification. Federal Railroad Administration.
3. Baumgardner, G., Sanchez-Badillo, A. (2024). Rail Crossing Violation Warning - Architecture and Design Specifications.
4. Zeiger, T., Baumgardner, G., Sanchez-Badillo, A. (2024) Rail Crossing Violation Warning Cloud Based Subsystem Bench Test & Evaluation Report
5. Sanchez-Badillo, A., Zeiger, T., Baumgardner, G (2025) RCVW C-V2X Field Test and Evaluation Report
6. Sanchez-Badillo, A., Baumgardner, G, Zeiger, T, Clark, N. (2024) RCVW Cloud-based System Demonstration Report
7. Sanchez-Badillo, A., Baumgardner, G., Paselsky, B., Seitz, T. (2022). “Appendix B. Vehicle-to-Infrastructure Rail Crossing Violation Warning – Phase II: Field Test & Evaluation Report.” Report No. DOT/FRA/ORD-22/07. Washington, DC: U.S. Department of Transportation, Federal Railroad Administration.
8. American Association of State and Highway Transportation Officials. (2018). A Policy on Geometric Design of Highways and Streets. AASHTO.

Abbreviations and Acronyms

ACRONYMS	EXPLANATION
AMPQ	Advanced Message Queuing Protocol
API	Application Programming Interface
BTE	Bench Test Evaluation
CBS	Cloud-based Subsystem
CICAS	Cooperative Intersection Collision Avoidance Systems
CORS	Constantly Operated Reference Station
ConOps	Concept of Operations
CP	Computer Platform
C-V2X	Cellular Vehicle-to-Everything
CV	Connected Vehicle
DCR	Data Communication Radio
DIO	Digital Input/Output
DOT	Department of Transportation
DSRC	Dedicated Short-Range Communications
DVI	Driver-Vehicle Interface
FCC	Federal Communications Commission
FDOT	Florida Department of Transportation
FRA	Federal Railroad Administration
FT&E	Field Test and Evaluation
GNSS	Global Navigation Satellite System
GUI	Graphical User Interface
HRI	Highway-Rail Intersection
HRIC	Highway-Rail Intersection Controller
ID	Identification Number
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
LTE	Long Term Evolution
MAP	HRI geometry message
NTRIP	Networked Transport of RTCM

OBU	On-Board Unit
OEM	Original Equipment Manufacturers
OS	Operating System
POE	Power Over Ethernet
RBS	Roadside-Based Subsystem
RCVW	Rail Crossing Violation Warning
RCVW-III	Rail Crossing Violation Warning Phase-III
RSU	Roadside Unit
RTCM	Radio Technical Commission for Maritime Services
RTK	Real-Time Kinematics
SAE	Society of Automotive Engineers
SDK	Software Development Kit
SPaT	Signal Phase and Timing
SRS	System Requirements Specification
TCP/IP	Transmission Control Protocol/Internet Protocol
TMC	Traffic Management Center
U.S. DOT	U.S. Department of Transportation
V2X	Vehicle to Everything
V2I	Vehicle and Infrastructure
V2V	Vehicle to Vehicle
VAC	Volts Alternate Current
VBS	Vehicle Based Subsystem
VDC	Volts Direct Current

Appendix A: Datasheets

[Neousys - Nuvo-7200VTC](#)

[Neousys - POC-451VTC](#)

[Danlaw - Autolink On-Board Unit](#)

[Danlaw - RouteLink Roadside Unit](#)

[Ublox – C102-F9R Application board](#)

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